Here are some hints or partial answers for problem set 04.

These were not proofread carefully, so please watch out for typographical or other errors.

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In this problem set, we consider only 'standard' Lorentz transformations, i.e, we only consider frames that are coincident at the common zero time, and have relative velocities in the common x direction.

Frame $\tilde{\Sigma}$ moves with velocity $v\hat{i}$ with respect to frame Σ. Here $\hat{i}$ is the unit vector along the common x, $\tilde{x}$ direction. Measurements of the same event from the two frames, (x, y, z, t) and $(\tilde{x}, \tilde{y}, \tilde{z}, \tilde{t})$, are related by the LT

$$\tilde{x} = \gamma_v(x - vt) \ ; \quad \tilde{y} = y \ ; \quad \tilde{z} = z \ ; \quad \tilde{t} = \gamma_v(t - vx/c^2) \ ;$$

or equivalently

$$\begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \gamma_v & -\gamma_v \frac{v}{c} & 0 & 0 \\ -\gamma_v \frac{v}{c} & \gamma_v & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix} \qquad \text{where } \gamma_v = (1 - v^2/c^2)^{-1/2}.$$

1. (a) Using the LT, derive algebraically the invariance

$$c^2\tilde{t}^2 - \tilde{x}^2 - \tilde{y}^2 - \tilde{z}^2 \ = \ c^2t^2 - x^2 - y^2 - z^2.$$

(In class we derived the LT from the invariance.)

(Partial) Solution/Hint →

$$\begin{aligned}
\tilde{x}^2 + \tilde{y}^2 + \tilde{z}^2 - c^2\tilde{t}^2 \ &= \ \gamma_v^2(x - vt)^2 + y^2 + z^2 - \gamma_v^2 c^2 (t - vx/c^2)^2 \\
&= \ \gamma_v^2 \left[\left(x^2 - 2xvt + v^2t^2\right) - c^2 \left(t^2 - 2xvt/c^2 + v^2x^2/c^2\right) \right] + y^2 + z^2 \\
&= \ \gamma_v^2 \left[(x^2 - c^2t^2) - \frac{v^2}{c^2}(x^2 - c^2t^2) \right] + y^2 + z^2 \\
&= \ \gamma_v^2 (x^2 - c^2t^2)(1 - v^2/c^2) + y^2 + z^2 \\
&= \ (x^2 - c^2t^2) + y^2 + z^2 \qquad \left\{ \text{since } \gamma_v^2 = \tfrac{1}{1 - v^2/c^2} \right. \\
&= \ x^2 + y^2 + z^2 - c^2t^2
\end{aligned}$$

which demonstrates the invariance.

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(b) **[1 pt.]** Write down the inverse LT, i.e., (x, y, z, t) in terms of $(\tilde{x}, \tilde{y}, \tilde{z}, \tilde{t})$, using the fact that frame Σ moves with velocity $-v\hat{i}$ with respect to frame $\tilde{\Sigma}$.

(Partial) Solution/Hint →

$$x = \gamma_v(\tilde{x} + v\tilde{t}) \ ; \quad y = \tilde{y} \ ; \quad z = \tilde{z} \ ; \quad t = \gamma_v(\tilde{t} + v\tilde{x}/c^2) \ .$$

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(c) Invert the LT algebraically to find the inverse LT, i.e., determine (x, y, z, t) in terms of $(\tilde{x}, \tilde{y}, \tilde{z}, \tilde{t})$ algebraically.

(Partial) Solution/Hint →

First get the perpendicular directions out of the way.

$$y = \tilde{y} \ ; \quad z = \tilde{z}$$

Now for x and t:

$$\begin{aligned} x = & \quad \frac{\tilde{x}}{\gamma_v} + vt && \text{from 1st } (\tilde{x}) \text{ eqn of LT} \\ = & \quad \frac{\tilde{x}}{\gamma_v} + v\left(\frac{\tilde{t}}{\gamma_v} + \frac{vx}{c^2}\right) && \text{using 4th } (\tilde{t}) \text{ eqn of LT} \\ = & \quad \frac{1}{\gamma_v}\left(\tilde{x} + v\tilde{t}\right) \ + \ \frac{v^2}{c^2}x \end{aligned}$$

$$\Longrightarrow \quad \left(1 - \frac{v^2}{c^2}\right)x \ = \ \frac{1}{\gamma_v}\left(\tilde{x} + v\tilde{t}\right) \qquad \Longrightarrow \quad \frac{x}{\gamma_v^2} \ = \ \frac{1}{\gamma_v}\left(\tilde{x} + v\tilde{t}\right)$$
$$\Longrightarrow \quad x \ = \ \gamma_v\left(\tilde{x} + v\tilde{t}\right)$$

We still need the equation for t. Using the 4th $(\tilde{t})$ eqn of LT,

$$\begin{aligned} t \ = \ \frac{\tilde{t}}{\gamma_v} + \frac{vx}{c^2} \ &= \ \frac{\tilde{t}}{\gamma_v} + \frac{v}{c^2}\gamma_v\left(\tilde{x} + v\tilde{t}\right) \ = \ \tilde{t}\left(\frac{1}{\gamma_v} + \gamma_v\frac{v^2}{c^2}\right) \ + \ \frac{v}{c^2}\gamma_v\tilde{x} \\ &= \ \gamma_v\tilde{t}\left(\frac{1}{\gamma_v^2} + \frac{v^2}{c^2}\right) + \gamma_v\frac{v}{c^2}\tilde{x} \ = \ \gamma_v\tilde{t} + \gamma_v\frac{v}{c^2}\tilde{x} \ = \ \gamma_v\left(\tilde{t} + \frac{v}{c^2}\tilde{x}\right) \end{aligned}$$

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(d) Two events have coordinates

$$(x_1, 0, 0, t_1) \quad \text{and} \quad (x_2, 0, 0, t_2)$$

in the Σ frame and

$$(\tilde{x}_1, 0, 0, \tilde{t}_1) \quad \text{and} \quad (\tilde{x}_2, 0, 0, \tilde{t}_2)$$

in the $\tilde{\Sigma}$ frame. If $\tilde{x}_1 = \tilde{x}_2 = \tilde{X}$, what is the proper time interval between the two events? Explain why.

Relate the time interval between the two events as measured in frame Σ with the time interval as measured in frame $\tilde{\Sigma}$.

(Partial) Solution/Hint →

Proper time in the $\tilde{\Sigma}$ frame, because that's the frame which measures them happening at the same place. i.e., proper time interval $= |\tilde{t}_1 = \tilde{t}_2|$.

Since time interval measured in any other frame is larger than the proper time interval,

$$|(t_1 - t_2)| \;=\; \gamma_v |(\tilde{t}_1 - \tilde{t}_2)|$$

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(e) In the Σ frame, the points A and B are fixed and have spatial coordinates

$$(x_A = L, y_A = 0, z_A = 0) \quad \text{and} \quad (x_B = 4L, y_B = 4L, z_B = 0)\ .$$

Find the time it takes for a light pulse emitted from point A to reach point B, as measured in the Σ frame and as measured in the $\tilde{\Sigma}$ frame. If you use the LT, specify clearly which events you are using it for.

Is the time measured in one of the frames a *proper* time interval? Explain, which frame, or why not.

(Partial) Solution/Hint →

In the Σ frame, the spatial distance between the two points is $\sqrt{(3L)^2 + (4L)^2} = 5L$, so the time required is

$$\tau = \frac{5L}{c}$$

Let's consider the events of light emission at A and light reaching B:

$$\begin{aligned} \text{1st event:} &\quad (x_1 = L, y_1 = 0, z_1 = 0, t_1) \\ \text{2nd event:} &\quad (x_2 = 4L, y_2 = 4L, z_2 = 0, t_2 = t_1 + \tau) \end{aligned}$$

In the $\tilde{\Sigma}$ frame, the two events are

$$\begin{aligned} \text{1st event:} &\quad \tilde{t}_1 = \gamma_v(t_1 - vx_1/c^2) = \gamma_v(t_1 - vL/c^2) \\ \text{2nd event:} &\quad \tilde{t}_2 = \gamma_v(t_2 - vx_2/c^2) = \gamma_v(t_2 - 4vL/c^2) \end{aligned}$$

so that the time taken in the $\tilde{\Sigma}$ frame is

$$\begin{aligned} \tilde{\tau} &= \tilde{t}_2 - \tilde{t}_1 = \gamma_v(t_2 - 4vL/c^2) - \gamma_v(t_1 - vL/c^2) \\ &= \gamma_v\left[(t_2 - t_1) - 3vL/c^2\right] \;=\; \gamma_v\left[\tau - 3vL/c^2\right] \\ &= \gamma_v\left[\frac{5L}{c} - \frac{3vL}{c^2}\right] \;=\; \frac{5L}{c}\gamma_v\left[1 - \frac{3}{5}\frac{v}{c}\right] \end{aligned}$$

In both frames, the events occur at different positions. So this is not a proper time interval in either frame.

This is consistent with the fact that the ratio between the time measured in the Σ frame and the that measured in the $\tilde{\Sigma}$ frame is not γ_v or $1/\gamma_v$.

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(f) In the Σ frame, two events occur at times $t_1 = \dfrac{L}{c}$ and $t_2 = \dfrac{L}{2c}$, and have spatial coordinates

$$(x_1 = 3L, y_1 = 0, z_1 = 0) \quad \text{and} \quad (x_2 = L, y_2 = d, z_2 = 0)$$

respectively. What is the speed v if the two events are found to be simultaneous in frame $\tilde{\Sigma}$?

(Partial) Solution/Hint →

The LT also holds for the difference of events, thus

$$\Delta\tilde{t} = \gamma_v(\Delta t - v\Delta x/c^2) \ .$$

If the events are simultaneous in frame $\tilde{\Sigma}$, we have $\Delta\tilde{t} = 0$, so that

$$\gamma_v(\Delta t - v\Delta x/c^2) = 0$$
$$\implies \frac{v}{c^2} = \frac{\Delta t}{\Delta x} = \frac{t_1 - t_2}{x_1 - x_2} = \frac{L/2c}{2L} = \frac{1}{4c} \qquad \implies \qquad v = c/4$$

Note that the difference of y coordinates ($y_1 = 0$ but $y_2 = d$) does not matter at all for the question asked.

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2. Let's omit the perpendicular directions (y, y', z, z').

   Using the rapidity $\phi = \tanh^{-1}(v/c)$, the LT can be written as

$$\begin{pmatrix} c\tilde{t} \\ \tilde{x} \end{pmatrix} = \begin{pmatrix} \cosh\phi & -\sinh\phi \\ -\sinh\phi & \cosh\phi \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix} = \Lambda(\phi) \begin{pmatrix} ct \\ x \end{pmatrix}$$

   (a) Show that two successive LT's, with rapidities ϕ_1 and ϕ_2, result in a LT of rapidtiy $\phi_1 + \phi_2$. In other words, show via matrix multiplication that

$$\Lambda(\phi_1)\Lambda(\phi_2) = \Lambda(\phi_1 + \phi_2)$$

   (Partial) Solution/Hint →

   Straightforward; I will not type this out.

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   (b) The velocities corresponding to rapidities ϕ_1 and ϕ_2 are v_1 and v_2 respectively. Find the velocity corresponding to $\phi_1 + \phi_2$, in terms of v_1 and v_2. How should we interpret this velocity?

   (Partial) Solution/Hint →

   The velocity corresponding to $\phi_1 + \phi_2$ is

$$v = c\tanh(\phi_1+\phi_2) = c\frac{\tanh\phi_1 + \tanh\phi_2}{1 + \tanh\phi_1 \tanh\phi_2} = c\frac{\frac{v_1}{c} + \frac{v_2}{c}}{1 + \frac{v_1}{c}\frac{v_2}{c}} = \frac{v_1 + v_2}{1 + \frac{v_1 v_2}{c^2}}$$

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(c) An LT at rapidity $-\phi$ should be the inverse of the LT at rapidity ϕ, i.e., we should have $\Lambda(-\phi) = [\Lambda(\phi)]^{-1}$. Show by matrix multiplication that $\Lambda(-\phi)\Lambda(\phi)$ is indeed the identity matrix.

(Partial) Solution/Hint →

$$\Lambda(\phi) = \begin{pmatrix} \cosh\phi & -\sinh\phi \\ -\sinh\phi & \cosh\phi \end{pmatrix} ;$$
$$\implies \Lambda(-\phi) = \begin{pmatrix} \cosh(-\phi) & -\sinh(-\phi) \\ -\sinh(-\phi) & \cosh(-\phi) \end{pmatrix} = \begin{pmatrix} \cosh\phi & \sinh\phi \\ \sinh\phi & \cosh\phi \end{pmatrix}$$

Therefore

$$\begin{aligned} \Lambda(-\phi)\Lambda(\phi) &= \begin{pmatrix} \cosh\phi & \sinh\phi \\ \sinh\phi & \cosh\phi \end{pmatrix} \begin{pmatrix} \cosh\phi & -\sinh\phi \\ -\sinh\phi & \cosh\phi \end{pmatrix} \\ &= \begin{pmatrix} \cosh^2\phi - \sinh^2\phi & 0 \\ 0 & \cosh^2\phi - \sinh^2\phi \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

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(d) Show that the set of Lorentz boosts in the x direction, represented by the matrices Λ, form a group under successive application (i.e., under matrix multiplication).

(Partial) Solution/Hint →

Group properties are satisfied:

Closure: multiplying two LT matrices gives another LT matrix, as shown above.

Associativity: Matrix multiplication is known to be associative.

Identity: The LT corresponding to zero velocity is an identity matrix, and hence an identity element for this group.

Inverse: For any LT, the LT corresponding to equal velocity in the opposite direction is the inverse element, as shown above.

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(e) Is this group (Lorentz boosts in the same direction) abelian or non-abelian? Explain why.

(Partial) Solution/Hint →

Since $\Lambda(\theta)\Lambda(\phi) = \Lambda(\theta + \phi) = \Lambda(\phi)\Lambda(\theta)$, multiplication of the LT matrices in the same direction is commutatiive. Hence the group is abelian.

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(f) Remember that rotation matrices are orthogonal. Check whether the Λ matrices are orthogonal. Explain why, or why not, in terms of an invariant.

(Partial) Solution/Hint →

$$\Lambda(\phi)^T = \begin{pmatrix} \cosh\phi & -\sinh\phi \\ -\sinh\phi & \cosh\phi \end{pmatrix}^T = \begin{pmatrix} \cosh\phi & -\sinh\phi \\ -\sinh\phi & \cosh\phi \end{pmatrix} = \Lambda(\phi)$$

Hence

$$\begin{aligned} \Lambda(\phi)^T\Lambda(\phi) \;=\; \Lambda(\phi)^2 \;&=\; \begin{pmatrix} \cosh\phi & -\sinh\phi \\ -\sinh\phi & \cosh\phi \end{pmatrix} \begin{pmatrix} \cosh\phi & -\sinh\phi \\ -\sinh\phi & \cosh\phi \end{pmatrix} \\ &=\; \begin{pmatrix} \cosh^2\phi + \sinh^2\phi & -2\cosh\phi\sinh\phi \\ -2\cosh\phi\sinh\phi & \cosh^2\phi + \sinh^2\phi \end{pmatrix} \end{aligned}$$

Since $\cosh^2\phi + \sinh^2\phi \neq 1$ and $\cosh\phi\sinh\phi \neq 0$ in general, this is not the identity matrix. Hence $\Lambda^T\Lambda \neq I$, i.e., Λ is not orthogonal.

In the case of rotation matrices, orthogonality follows from the preservation of Euclidean norm (magnitude of the position vector) under transformations. In the case of the LT, the corresponding quantity $(ct)^2 + x^2$, is not preserved. Hence, the transformation matrices are not expected to be orthogonal.

In fact, the quantity preserved is $(ct)^2 - x^2$, which leads to the following analog of the orthogonality condition:

$$\Lambda^T g \Lambda = I$$

where g is the matrix representing the metric tensor; in the present 2×2 case it is $g = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

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(g) Remember that rotation matrices have unit determinant. Check whether the Lorentz boost matrices Λ have unit determinant.

(Partial) Solution/Hint →

$$\det\left[\Lambda(\phi)\right] \;=\; \begin{vmatrix} \cosh\phi & -\sinh\phi \\ -\sinh\phi & \cosh\phi \end{vmatrix} = \cosh^2\phi - \sinh^2\phi \;=\; 1$$

Thus LT matrices have unit determinant. LT matrices are examples of matrices that have unit determinant but are not orthogonal.

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